

Batch 1 (Physics-A): Units + motion + force + energy + heat

CONCEPT	FORMULA / RULE	TRAP
SI base + derived	m/kg/s/A/K/mol/cd + N/J/W/V/ohm; N = kg m per s^2.	Unit without symbol/formula link.
Prefixes + notation	kilo/milli/micro/nano/pico + powers; 3.2 x 10^5 conversions.	Power/direction dropped.
Scalar vs vector	Scalar magnitude only (speed/work); vector + direction (velocity/force).	Direction ignored.
Distance vs displacement	Path scalar vs straight-line vector; return trip gives zero displacement.	Path = displacement assumed.
Speed/velocity/accel	v = s/t; a = change-v/t; m-per-s and m-per-s^2 units.	Direction/unit dropped.
Graphs	Distance-time slope = speed; velocity-time slope = accel, area = displacement.	Slope/area unread.
Motion + force	v = u+at; s = ut+half a t^2; v^2 = u^2+2as; F = m a net.	Units/net-force skipped.
Newton 1 + 2	Inertia (seat belt); F = ma, heavier needs bigger F.	Example/net missed.
Newton 3 + momentum	Equal-opposite pairs (rocket); p = mv; impulse = F t.	Single-body reaction.
Work + power	W = F d along force (zero if perpendicular); P = W/t; joule/watt.	Angle/unit ignored.
KE + PE	KE = half m v^2 (double v = 4x); PE = m g h.	Squared factor missed.
Conservation + heat	Total constant, KE/PE swap; heat = energy (joule), temp = degree (C/F/K, 0C = 32F = 273K).	Lost-not-converted; heat = temp.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Unit + symbol + link trio?	A: What + symbol + formula
Q: Double v does what to KE?	A: 4x
Q: 0 C in F and K?	A: 32 F; 273 K

Batch 2 (Physics-B): Transfer + light + sound + circuits start

CONCEPT	FORMULA / RULE	TRAP
Conduction/convection	Solids-touch-metals vs fluids-bulk-flow (spoon vs sea breeze).	Medium missed.
Radiation/expansion	No-medium waves (sun); heat expands (rail gaps, bimetal).	Medium/gap missed.
Heat capacity/latent	$Q = m c \text{ change-T}$, water highest; flat at melt/boil (steam worse).	Flat/capacity missed.
Reflection + plane	Angle-in = angle-out; virtual erect same-size lateral-flip.	Virtual/flip missed.
Spherical + refraction	Concave converges, convex diverges; bend at boundary (straw/pool).	Converge/medium missed.
Lenses + eye	Convex converges, concave diverges; lens focuses, retina senses.	Converge/retina missed.
Defects + scopes	Myopia far-blur concave; hyper near-blur convex; scopes magnify.	Near/far + lens swapped.
Sound + pitch	Mechanical, needs medium (none in vacuum); high-f = shrill.	Medium/pitch missed.
Loudness + $v=fL$	Amplitude = loud; $v = f \times \text{wavelength}$; double f halves L.	Amplitude/product missed.
Ranges + ultrasound	Infra below 20, audible 20-20000, ultra above; scan/clean/sonar.	Range/use missed.
Echo + charge	17 m minimum; $Q = I \times t$, coulomb.	Distance/flow missed.
Voltage + Ohm	Push in volts; $V = I R$; double V doubles I.	Push/ratio missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Slope vs area on graphs?	A: Slope = rate; area = quantity
Q: Double f does what to L?	A: Halves it
Q: Ultrasound uses?	A: Scan/clean/sonar

Batch 3 (Circuits + Chem-A): Resistors + power + atom + table + pH

CONCEPT	FORMULA / RULE	TRAP
Series/parallel	$R_s = R_1 + R_2$ same-current; same-voltage with $1/R_p$ sum; home = parallel.	Mode/same missed.
Power + kWh	$P = V \times I$; units kWh = bill units (2 kW x 3 h = 6).	Formula/kWh missed.
Conductors/magnets	Copper flows, rubber blocks; N/S like-repels; electromagnet = coil + current + soft core.	Poles/current missed.
Safety + states	Fuse melts, MCB trips, earth guards; solid fixed/liquid flows/gas fills.	Device/state missed.
Changes + types	Melt/freeze/boil/sublime; physical reversible vs chemical new (burn/rust).	Reverse/new missed.
Atom + matter	Atom/molecule/charged-ion; element pure-one, compound fixed-ratio, mixture separable (air).	Ratio/separable missed.
Mixtures + p/n/e	Homogeneous uniform vs hetero; proton +/neutron 0/electron -.	Uniform/charge missed.
Z/A + isotopes	Z = protons, A = p+n; same-Z different-neutrons (C-12/14, U-235).	Z/same missed.
Periodic + acid/base	Periods rows, groups columns, metals left; acid H+ sour, base OH- soapy.	Trend/ion missed.
pH + indicators	Below 7 acid, 7 neutral, above base (blood 7.4); litmus/phenol/methyl flips.	Number/flip missed.
Neutralize + acids	Acid + base = salt + water (antacid); HCl/H ₂ SO ₄ /HNO ₃ + NaOH/CaO.	Product/name missed.
Common salts	NaCl table, Na ₂ CO ₃ washing, NaHCO ₃ baking, POP plaster.	Formula/use missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Series vs parallel same-thing?	A: Current vs voltage
Q: pH of blood?	A: 7.4 (mild base)
Q: Antacid chemistry?	A: Neutralization: salt + water

Batch 4 (Chem-B + Bio-A): Daily chem + fire + cell organelles

CONCEPT	FORMULA / RULE	TRAP
Daily pH + metals	Stomach acid/antacid, soil/rain; lustre/malleable/ductile/conduct.	Life/trait missed.
Reactivity series	K Na Ca Mg Al Zn Fe Pb H Cu; upper displaces lower (Zn + CuSO ₄).	Order/displace missed.
Rust + shield	Iron + air + water = rust; oil/paint/galvanize/alloy blocks.	Needs/shield missed.
Alloys + uses	Brass/bronze/steel; Cu wires, Al foil, graphite + S tyres.	Mix/use missed.
Soap vs detergent	Soap = fat + alkali, fails in hard water; detergent works.	Hard-water fail missed.
Bleach/fertilizer/fuel	Chlorine whitens; NPK + urea/potash feed; petrol/diesel/CNG burn.	Agent/NPK/burn missed.
Polymer + home	Repeat units (polythene/PVC/tyres); cleaners/phenyl/naphthalene.	Repeat/daily missed.
Hard water + fire	Ca/Mg scum + scale (boil/ion-exchange); fuel + oxygen + heat, remove one.	Ions/legs missed.
Theory + prokaryote	Cell = unit, all from cells; prokaryote no-nucleus (E. coli).	Unit/nucleus missed.
Plant/animal + gate	Wall + big vacuole + chloroplast vs none; membrane selective, wall rigid.	Wall/gate missed.
Nucleus + mito	DNA control (brain); ATP powerhouse (muscle-rich).	DNA/ATP missed.
Ribosome + ER/Golgi	Protein factory; ER road, Golgi post-office packs/ships.	Make/pack missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Displace rule + example?	A: Upper displaces lower; Zn + CuSO ₄
Q: Fire triangle legs?	A: Fuel + oxygen + heat
Q: Powerhouse + brain of cell?	A: Mitochondria; nucleus

Batch 5 (Bio systems): Division + systems + nutrients + minerals

CONCEPT	FORMULA / RULE	TRAP
Lysosome + chloro	Suicide-bag digests, vacuole stores; chlorophyll cooks (plants only).	Digest/pigment missed.
Mitosis/meiosis	1-into-2 equal (growth/repair) vs halve + gametes + variation.	Equal/halve missed.
Digestive order	Mouth/foodpipe/stomach/intestine + liver/pancreas; villi absorb.	Gland missed.
Lungs + heart	Alveoli O ₂ /CO ₂ exchange; double circulation, arteries/veins, RBC carry.	Exchange/circuit missed.
Nervous + endocrine	Neuron/reflex fast (knee jerk) vs pituitary/thyroid/insulin slow.	Speed/gland missed.
Kidney + largest	Nephron filters urine; skin largest organ, liver gland, femur bone.	Unit/trap missed.
Macros + fibre	Carbs energy, protein growth, fat reserve; water solvent + fibre roughage.	Role/rough missed.
A + B vitamins	A night vision/retinol/xerophthalmia; B1 beriberi, B12 anaemia.	Disease/source missed.
C + D	C citrus/scurvy/heal; D sun/rickets/bones + milk.	Disease/source missed.
E/K + split + iron	E protects, K clots; fat = A/D/E/K, water = B/C; iron = Hb/anaemia.	Split/iron missed.
Iodine + calcium	Thyroid/goitre/iodized salt; bones/teeth + milk/ragi.	Disease/source missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Equal vs halve division?	A: Mitosis; meiosis
Q: Fat vs water vitamins?	A: A/D/E/K; B/C
Q: Goitre + bone minerals?	A: Iodine; calcium

Batch 6 (Disease + plants): Immunity + photo + hormones + food chains

CONCEPT	FORMULA / RULE	TRAP
Spread + germs	Communicable spreads vs lifestyle NCD; virus/bacteria/fungus/protozoa.	Spread/type missed.
Vectors + routes	Mosquito/fly/rat carry; air/water/food/contact spread.	Carrier/route missed.
Immunity + flags	Innate born vs acquired built; antigen flags, antibody fights.	Born/flag missed.
Vaccine + drugs	Weak/killed trains (BCG/polio); antibiotics bacteria-only, full course.	Prevent/virus-limit missed.
Prevent + photo	Hygiene/vaccine/nets; CO ₂ + water + sun + chlorophyll = glucose.	Prevent/equation missed.
Factors + loss	Light/CO ₂ /water/temp set rate; stomata loss pulls water up.	Factor/loss missed.
Pipes + flower	Xylem water-up, phloem food-both-ways; pollinate-fertilize-fruit-seed.	Direction/flower missed.
Clone + auxin	Runner/bud/graft/tissue same-copy; tip auxin bends stem to light.	Clone/bend missed.
Hormones + moves	Gibberellin stem, cytokinin divide, ABA sleep, ethylene ripen; photo/geo/hydro tropisms.	Role/move missed.
Eco + chain	Living + nonliving interact (pond); linear links grass-deer-lion.	Unit/link missed.
Web + trophic	Many-chain net = stable; 10% energy up levels.	Net/percent missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Born vs built defence?	A: Innate; acquired
Q: Xylem vs phloem?	A: Water up; food both ways
Q: Energy passed up levels?	A: 10%

Batch 7 (Env + scientists): Pollution + gases + waste + names

CONCEPT	FORMULA / RULE	TRAP
Pyramid + bio	Number/biomass/energy shapes; genes/species/eco variety + hotspots (W. Ghats).	Type/variety missed.
Conserve + air	In-situ home (Kaziranga) vs ex-situ away (zoo/seed bank); SPM/RSPM + smog.	Home/pollutant missed.
Water + blanket	Sewage/industry + BOD (Ganga); greenhouse blanket warms.	BOD/blanket missed.
Gases + ozone	CO ₂ /CH ₄ /N ₂ O/CFC + sources; ozone blocks UV, CFC hole, Montreal.	List/shield missed.
Acid + waste	SO ₂ /NO _x sour rain (Taj); reduce/reuse/recycle + segregate.	Gas/order missed.
Newton + Ohm	3 motion laws + gravity; $V = IR$ + resistance.	Work/formula missed.
Faraday + Rutherford	Induction + EMF (motor/generator); gold-foil nucleus.	Work/model missed.
Thomson + Chadwick	Electron + plum-pudding (cathode rays); neutral neutron.	Particle/model missed.
Mendeleev + Lavoisier	Periodic table + gaps predicted; oxygen + combustion + mass conserved.	Predict missed.
Pasteur + Jenner	Germs + pasteurize + rabies; smallpox first vaccine (UIP link).	Work/disease missed.
Koch + Darwin	Postulates + TB/cholera proof; selection + evolve (finches).	Proof/select missed.

60-SECOND SPEED ROUND (cover answers, recall aloud)

Q: Home vs away conservation?	A: In-situ; ex-situ
Q: Ozone hole cause + pact?	A: CFC; Montreal
Q: First vaccine + disease?	A: Jenner; smallpox